8 The equation of a curve is $y = 5e^{-\frac{1}{2}x} + e^{\frac{1}{2}x}$.

(a) Find the coordinates of the stationary point, in logarithmic and/or surd from where appropriate. [5]

$y = 5e^{-\frac{1}{2}x} + e^{\frac{1}{2}x}$

$\frac{dy}{dx} = 5\left[e^{-\frac{1}{2}x}\right]\left[-\frac{1}{2}\right] + e^{\frac{1}{2}x}\left(\frac{1}{2}\right)$

$= -\frac{5}{2}e^{-\frac{1}{2}x} + \frac{1}{2}e^{\frac{1}{2}x}$ M1

for stationary pts,

$\frac{dy}{dx} = 0$

$\frac{5}{2}e^{-\frac{1}{2}x} = \frac{1}{2}e^{\frac{1}{2}x}$ M1

$\times e^{\frac{1}{2}x}$, $\frac{5}{2}e^{0} = \frac{1}{2}e^{x}$

$e^{x} = 5$

$\ln e^{x} = \ln 5$

$x = \ln 5$ M1

$y = 5e^{-\frac{1}{2}\ln 5} + e^{\frac{1}{2}\ln 5}$

$= 5e^{\ln 5^{-1/2}} + e^{\ln 5^{1/2}}$

$= 5\left[5^{-1/2}\right] + 5^{\frac{1}{2}}$

$= 5^{\frac{1}{2}} + 5^{\frac{1}{2}}$

$= 2\sqrt{5}$ M1

$\therefore$ stationary pt is $(\ln 5, 2\sqrt{5})$ A1

$a^{\log_a b} = b$

$x = a^{\log_a b}$

$\log_a x = \log_a a^{\log_a b}$

$\log_a x = \log_a b$

$\therefore x = b$

(b) Determine the nature of the stationary point of the curve. [2]

$$\frac{dy}{dx} = -\frac{5}{2}e^{-\frac{1}{2}x} + \frac{1}{2}e^{\frac{1}{2}x}$$

$$\frac{d^2y}{dx^2} = -\frac{5}{2}\left[e^{-\frac{1}{2}x}\right]\left[-\frac{1}{2}\right] + \frac{1}{2}\left[e^{\frac{1}{2}x}\right]\left[\frac{1}{2}\right]$$

$$= +\frac{5}{4}e^{-\frac{1}{2}x} + \frac{1}{4}e^{\frac{1}{2}x} \quad \text{M1}$$

When $x = \ln 5$,

$$\frac{d^2y}{dx^2} > 0$$

$\Rightarrow (\ln 5, 2\sqrt{5})$ is a minimum point. A1

(c) Explain why the gradient $\frac{dy}{dx}$ is an increasing function for all real values of x. [2]

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2} = +\frac{5}{4}e^{-\frac{1}{2}x} + \frac{1}{4}e^{\frac{1}{2}x}$$

Since $e^{-\frac{1}{2}x} > 0$, $e^{\frac{1}{2}x} > 0$ M1

$\Rightarrow +\frac{5}{4}e^{-\frac{1}{2}x} > 0$ $\frac{1}{4}e^{\frac{1}{2}x} > 0$

$\therefore \frac{d^2y}{dx^2} > 0$

A1

$\Rightarrow$ gradient is an increasing function for all real values of x. (explained)